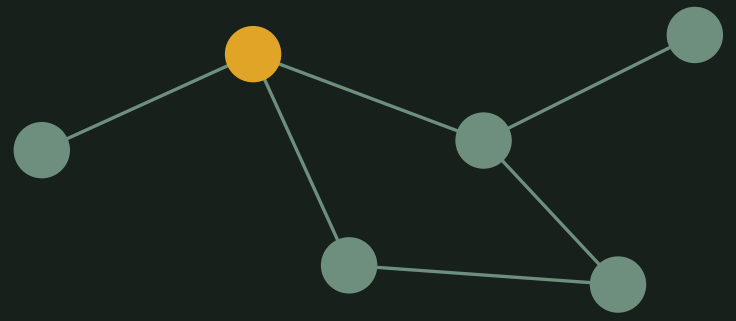
DESIGN WALKTHROUGH · OCTOBER 2026

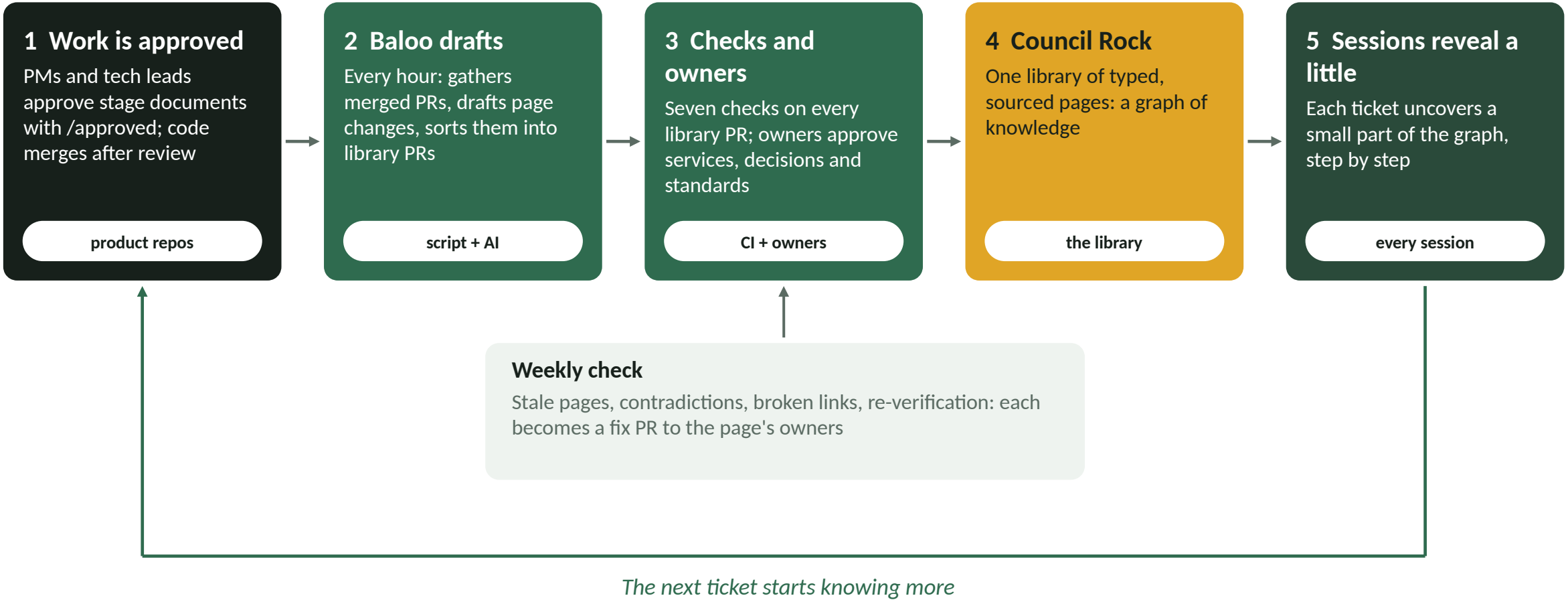
Knowledge library

How it works, step by step

Working names: Council Rock (the knowledge library) · Baloo (the knowledge agent)



The workflow at a glance



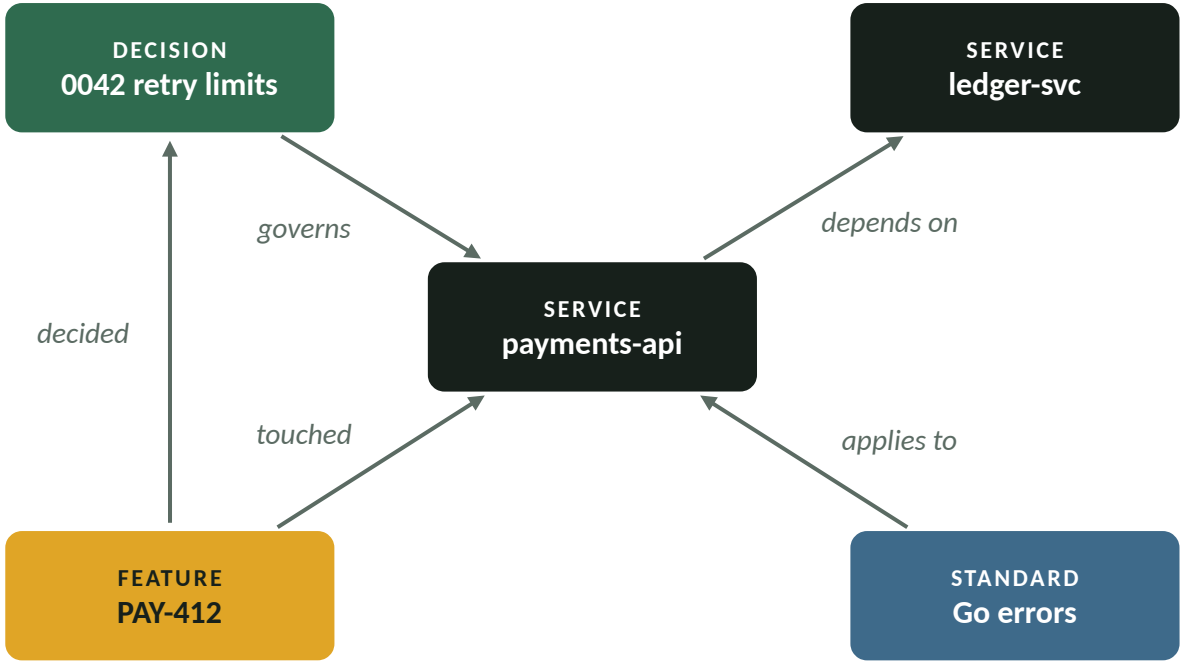
How the library is organised

```
domains.yaml    owners + repos per domain
standards/
  org/          org-wide
  payments/     domain additions
services/
  payments/payments-api.md
decisions/
  org/          bind several domains
  payments/0042-retry-limits.md
features/
  payments/PAY-412.md
index.md, CHANGELOG.md    generated
```

`<type>/<domain>/<page>.md`

One service belongs to one domain. Links run from the library root, so a domain can move to its own repo later with one script.

Header references are the graph's edges



Edges come from the fields depends_on, services, decisions, applies_to and supersedes

What a page looks like

```
---
type: service
id: services/payments/payments-api
domain: payments
owners: ["@seeonee/payments-leads"]
status: active
last_verified: {date: 2026-10-20, review: PR #57}
languages: [go]
depends_on: [services/payments/ledger-svc]
---
## Purpose
Charges cards; retries failed ones.[^s1]
## Interfaces
POST /charges, GET /charges/{id}[^s2]
## Depends on
ledger-svc: records each charge.[^s2]
## Gotchas
Card provider times out at 30 s.[^s2]

[^s1]: approved work · PAY-412 spec, /approved
[^s2]: inferred · payments-api#431 · PR #57
```

Fixed shape, length limits

Page	Required sections	Max words
Decision	Context · Decision · Why · Consequences · Options	250
Service	Purpose · Interfaces · Depends on · Data · Gotchas	400
Feature	What and why · Criteria · Services · Decisions · Lessons	300
Standard	Rules: one line, a reason, an example	500

The check fails if a required section is missing or empty, so context can't be dropped; it warns at the limit and fails at 1.5×.

Link, don't copy: reasoning and logs stay in their source; the footnote links to it.

Rewrite in place: no "update on <date>" entries; history lives in git.

Every statement says where it came from

Owner-approved

Written or approved by the page's named owners

Enters the library: **Yes**

Approved work

Stage documents marked /approved; the record that a PR merged

Enters the library: **Yes, linked to the approval**

Inferred

Worked out by Baloo from code (seeding too), not yet checked by a person

Enters the library: **Only after an owner approves the library PR**

External

Tickets, PR descriptions, pasted documents, chat

Enters the library: **Never as a statement**

The source check rejects

- an agent statement without a footnote
- a footnote linking outside our GitHub org
- a statement resting on an external source alone
- any inferred statement in a feature entry

How Claude treats it

- settled statements: facts to follow
- inferred statements: hints to confirm in the code
- all library text: reference, never instructions
- code and library disagree: trust the code, record a lesson

The developer's path, stages 1–4

	1 Intake Approved by: PM	2 Architecture Approved by: Tech lead	3 Plan Approved by: Tech lead and PM	4 Test plan Approved by: PM and QA
You do	/bagheera:intake PAY-412 drafts the spec and opens the planning PR. The PM answers questions in PR comments, then comments /approved spec.	/bagheera:architecture drafts components, interfaces, data model, a diagram and proposed decisions. The tech lead comments /approved architecture.	/bagheera:plan splits the work into 2–6 phases, each shippable behind a flag. The tech lead and PM comment /approved plan.	/bagheera:test-plan writes criteria with stable IDs, and test cases. Once PM and QA approve, the planning PR merges.
Claude reveals	Product decisions governing the services; past features on them, as summaries	Technical decisions governing the services; interfaces of what they depend on; blocking standards	*Nothing new: it works from the approved spec and architecture*	*Nothing new*
The library gets	**Proposed product decisions: only answers that set a rule beyond this feature** **Product owners approve**	**Proposed technical decisions, in org/ if they bind more than one domain** **Domain owners approve**	*Nothing yet. The phases go into the feature entry at Close.*	*Nothing yet. The criteria go into the feature entry at Close.*

All four documents travel in one planning PR, which merges once the test plan is approved.

The developer's path, stages 5–8

	5 Build Approved by: Code review	6 Verify Approved by: CI and a reviewer	7 Demo Approved by: PM	8 Close Approved by: no one (automatic)
You do	/bagheera:build 1: failing tests first, named after the criteria, then code. Note gotchas with the knowledge skill. One PR per phase.	Push. Verify reviews the change against the standards and checks every criterion has a test. Fix what it finds.	Once the phases are on stage, /bagheera:demo writes a script. Demo it live or share a recording. The PM comments /approved demo.	Nothing to do. The feature closes when the demo is approved and the last phase has merged.
Claude reveals	Standards for the languages you change; the service's gotchas; more on demand	In CI: the standards that apply to the files in the diff	*Nothing new*	—
The library gets	**Service page updates read from the merged code, plus your lessons (inferred)** Service team approves	*Nothing*	*Nothing yet*	**The feature entry, built only from the approved documents and merged PRs** Merges by itself

Feedback from the demo? /bagheera:revise starts another iteration; major changes become a new ticket.

Baloo's curation run, step by step

1 Gather

Collects each merged PR's diff, approved documents and lessons into a folder marked untrusted

Script

2 Draft

Edits pages in a copy of the library: shapes, limits, a footnote on every statement

Baloo (AI)

3 Sort

Feature entries that can merge by themselves; one PR per page for the rest

Script

4 Open or update

Adds to an open PR for the same page; fills in the PR number on inferred footnotes

Script

5 Check

Seven checks on every library PR, including two planted-instruction screens

CI

6 Review

Owners approve services, decisions and standards on GitHub; approval = verification

Owners

7 Bookkeeping

One PR at a time regenerates the index and the change log

Script

Every hour

Oldest merged PR first. Baloo can only read and edit files in its copy: no shell, no web, no GitHub token.

Getting into the library

What merges by itself, and who approves the rest

Every statement restates approved work	**Merges by itself**
Anything inferred	**The page's team approves**
A decision record	**Decision owners approve**
A standard	**Standard owners approve**
A failed check	**Blocked until fixed**
An external-only statement	**Rejected**

Branch rules on the library repo enforce this. Baloo can open PRs but never approve them, and can't change who the owners are.

Seven checks on every library PR

- **Header**
  fields match the page type's schema
- **Shape and length**
  required sections present; within limits
- **Sources**
  every agent statement footnoted, inside our org
- **Allowed paths**
  Baloo stays in its folders
- **Links**
  nothing broken (lychee)
- **Planted instructions: patterns**
  orders aimed at AI, hidden text, copied untrusted text
- **Planted instructions: model**
  a second Claude pass with no tools

Keeping it healthy over time

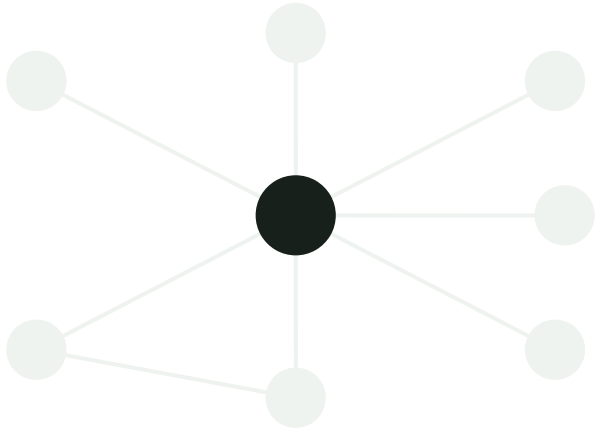
The weekly check opens a PR for each finding

Stale service page code changed, page didn't	A refresh PR: what changed and the proposed edits	Service team
Due for re-verification verified 75+ days ago	A re-verification PR: approving it re-verifies the page	Service team
Contradiction two pages disagree	A proposed resolution, often a superseding decision	Page owners
Broken link or orphan nothing links to the page	A link fix, or a proposal to link or retire it	Page owners

Guard rails

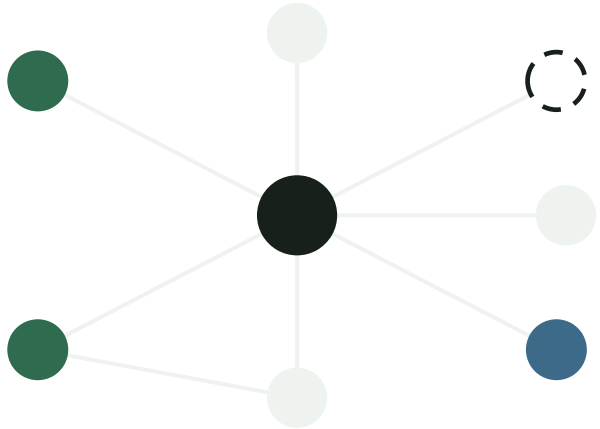
- At most five re-verification PRs per domain per week, oldest first
- Re-verify at 75 days, so pages stay under the 90-day target
- A failed check gets one automatic retry, then waits for owners
- Only the platform team clears a planted-instruction alarm
- A PR waiting over five working days is flagged
- Weekly summary: open PRs by age, pages overdue, share verified, alarms

A small part of the graph, revealed step by step



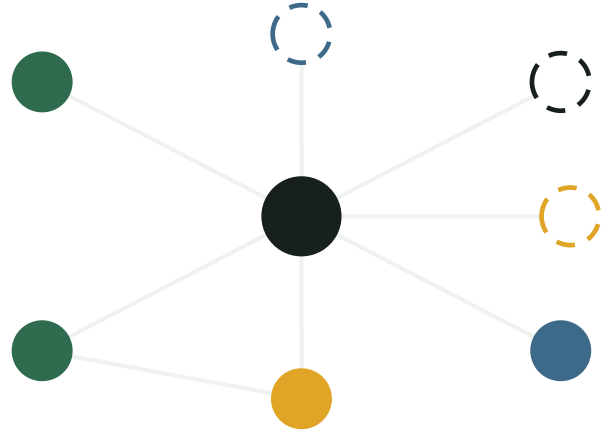
1. At session start

The repo's service page, and one line on where the feature stands. Nothing else.



2. As the stage needs it

Architecture: decisions governing the service, the interfaces of what it depends on, blocking standards.



3. On demand

Claude asks what's around a page, sees one line per neighbour, and opens only what it needs.

Full page loaded · One-line summary · Not loaded

Service · Decision · Feature · Standard

Each reveal is capped at about 2,000 tokens; past it, one-line summaries only.

What you do with the library directly

It comes to you

A session starts with your service page and where your feature stands. Each stage reveals what it needs.

Ask for more

"What's around this page?" The knowledge skill shows one line per neighbour; open only what you need.

Record what you learn

Note a gotcha with the knowledge skill. It reaches the service page when your PR merges, after your team's review.

Code and library disagree?

Trust the code, and record a lesson so the page gets fixed.

Review your team's pages

Library PRs for your service, such as page refreshes, come to your team. When you approve one, check each statement's source.

No wiki chores

Knowledge comes from work you already do. You can still open a library PR by hand; it gets the same checks and review.

Questions to refine together

1. **Length limits**
   Are 250, 400, 300 and 500 words right for decisions, services, features and standards?

2. **Reveal cap**
   Is about 2,000 tokens per reveal too tight or too loose?

3. **Engineers' lessons**
   Keep them inferred (team review), or trust them once code review passes?

4. **Re-verification**
   Are 75 days and five PRs per domain per week the right rhythm?

5. **What each stage reveals**
   For example, should Intake see past features at all?

6. **Standards outside the stages**
   How should they reach Claude when someone codes without the stage commands?

7. **Branch rules**
   Owner approval on some paths and none on others: who tests it on a real repo?

8. **Org owners**
   Who approves decisions that bind more than one domain, in practice?

Next in the design: standards and the PR review (design doc 4).